

1 Numerical integration on the Lorenz system

1. Integrate numerically the system

$$\begin{cases} \dot{x} = \sigma(y - x) \\ \dot{y} = rx - xz - y \\ \dot{z} = xy - bz \end{cases} \quad (1)$$

up to $t=60$, with a time step $dt = 0.005$ using the Euler forward method, with the following sets of values for the parameters

- Set A: $\sigma, b, r = (10, 8/3, 28)$
- Set B: $\sigma, b, r = (10, 8/3, 9)$

and the initial condition $L_0 \equiv (x_0, y_0, z_0) = (9, 10, 18)$. Plot trajectories $L(t, L_0)$ in the (x, z) space and describe the result (max 100 words).

2. Repeat 1. with $L'_0 = (x'_0, y'_0, z'_0) = (9 + \epsilon, 10, 18)$ with $\epsilon = 1 \times 10^{-7}$. Plot $x(t)$ for the two solutions $L(t, L_0)$ and $L(t, L'_0)$ and their difference. Describe the result (max 100 words).

3. For the set of parameters A, treat $L(t, L_0)$ as an observed state and $L(t, L'_0) = (x', y', z')$ as an integration affected by an error ϵ in the initial condition. Compute the *root-mean-square error* (RMSE) for the x variable

$$R(t) \equiv \sqrt{(x(t) - x'(t))^2} = \sqrt{(x_{true}(t) - x'(t))^2}$$

as function of lead time and plot it on a linear and semilog scale. Describe and discuss the result (max 150 words).

4. Generate an *ensemble of initial conditions* by perturbing the initial condition L_0 with a random perturbation $E_k \equiv (\epsilon_k, 0, 0)$ for $k = 1, \dots, N$ and $N=100$. Suppose for simplicity that ϵ_k is uniformly distributed between -0.75 and 0.75. Note that `numpy.random.random()` generates a random real number between 0 and 1. For each member of the ensemble $L'_k(t, L_0 + E_k)$ integrate the system up to $t=4$ and compute the mean square error

$$R_k^2(t) = (x_{true}(t) - x_k(t))^2$$

and the mean square error of the average $\langle x \rangle$, where

$$\langle \dots \rangle = \frac{1}{N} \sum_{k=1}^N (\dots).$$

Compare the latter with the average mean square error $\langle R^2 \rangle = \frac{1}{N} \sum_{k=1}^N R_k^2$ by plotting them as function of lead time. Describe and discuss the result.

5. (OPTIONAL) Integrate numerically the system

$$\begin{cases} \dot{x} = \sigma(y - x) \\ \dot{y} = rx - xz - y + f(t) \\ \dot{z} = xy - bz \end{cases} \quad (2)$$

with $f(t) = 0$ and with an ensemble of 50 initial conditions with a uniform distribution of x between -20 and 20, y between -20 and 20, and z between 0 and 30, up to $t=200$. For each member k , consider subsets of the time series each 5000 time step long, from $5000 \cdot \tau$ to $5000 \cdot (\tau + 1)$ with $\tau = 0, 1, 2, \dots, M$ (choose the maximum M given the length of the time series). Define the probability p_x that $x < 0$ and estimate it in each chunk of 5000 time steps, compute this value for each τ and assign it to an array $P_k(\tau)$. Plot $P_k(\tau)$ for each member and plot the mean $\langle P_k \rangle$. Repeat all these tasks with $f(t) = \sqrt{t}$. Plot also $f(t)$. Describe the result and discuss what we can conclude about the interpretation of the response of a chaotic system to a forcing f (max 200 words).

6. Write a short report, including a brief introduction, figures and text produced during the session (points 1-4 and 5 where applicable) and a conclusive comment (max 4 pages).